Cloud Deployment Models

Task-1.Pick any two apps you use daily (like Zomato, Instagram, or Paytm) and identify which cloud deployment model (Public, Private, Hybrid, or Community) would best suit each app if you were the CTO. Justify your choice with two reasons for each.

App	Best deployment model	Reasons
Instagram	Public Cloud	1. It needs to serve millions/billions of users globally and can scale resources according to demand. 2. Public cloud infrastructure provides worldwide availability without requiring the company to build every data center itself.
Paytm	Hybrid Cloud	1. Sensitive financial/customer information can be kept in tightly controlled infrastructure. 2. Public-cloud resources can be used for high-traffic services and scalability during peak periods such as sales or bill-payment deadlines.

Task-2.Create a comparison table listing the pros and cons of Public, Private, Hybrid, and Community cloud deployment models. Include at least three points each for pros and cons.

Deployment Model	Pros	Cons
Public Cloud	• Lower infrastructure cost• Highly scalable• Easy global access	• Less direct control over physical infrastructure• Ongoing cloud costs can increase with usage• Requires careful configuration for security
Private Cloud	• Greater control over infrastructure• Strong customization options• Suitable for sensitive workloads	• Expensive to build and maintain• Requires skilled IT staff• Scaling can be slower/costlier
Hybrid Cloud	• Combines flexibility of public cloud with private infrastructure• Sensitive data can remain private• Good scalability for variable workloads	• More complex to manage• Integration between environments can be difficult• Security policies

Deployment Model	Pros	Cons
		must be managed across multiple environments
Community Cloud	• Shared by organizations with similar requirements• Costs can be shared• Can support common security/compliance requirements	• Smaller user base than public cloud• Shared governance can be complicated• Less flexibility than a dedicated private cloud

Task-3.Read the case study of WhatsApp's cloud infrastructure (search online for details) and summarize which deployment model they use, explaining why this model is suitable for their business needs.Focus on scalability, security, and global reach.

Deployment model: Primarily Private Cloud / company-owned infrastructure

WhatsApp is operated by Meta, which builds and operates its own large-scale data-center infrastructure. Meta says it has 32 owned and operated data centers supporting its apps and technologies, including WhatsApp.

This model is suitable because:

- **Scalability: Meta can design and expand its infrastructure specifically for the enormous workload generated by WhatsApp and its other products. Meta describes its infrastructure as a globally networked operation serving billions of people.**
- **Security: WhatsApp uses end-to-end encryption, and Meta has built specialized security infrastructure for features such as encrypted backups and private AI processing.**
- **Global reach: Meta operates infrastructure across multiple geographic locations, allowing its services to support users around the world. Meta is also expanding infrastructure in India, including an AI-enabled data center in Jamnagar, Gujarat.**

Note: In modern practice, Meta's infrastructure strategy is more accurately described as a mix of owned infrastructure and cloud-provider partnerships, rather than a textbook single deployment model. For example, Meta announced a large AWS Graviton deployment in 2026.

Task-4.Imagine you are launching a new online ticket booking platform for IPL matches. Choose a cloud deployment model for your platform, list two possible risks with your choice, and suggest a way to mitigate one of those risks

Chosen model: Public Cloud

I would choose a Public Cloud because IPL ticket sales can experience enormous and unpredictable traffic spikes when tickets become available.

Two possible risks:

1. **Service overload: A sudden rush of users could cause slow response times or downtime.**
2. **Security risks: Attackers could attempt DDoS attacks, account attacks, or other attacks during high-demand ticket releases.**

Mitigation for service overload:
Use auto-scaling and load balancing so that additional servers/resources are automatically added when traffic increases and removed when demand decreases. A CDN and caching can also reduce pressure on application servers.

This makes the public cloud a good choice because it provides the flexibility to handle normal traffic as well as sudden IPL ticket-booking spikes.